Devon,

Thank you for the opportunity to help support Summit Telecom's outreach to increase retention for the company. In this project, our goal is to create a list for your team to work off of, identifying customers with the highest probability of churn, as FTE capacity allows for 20% of customers to receive personal outreach. It cannot establish the effect of contacting people.

After evaluating three models: Contract Rule, Logistical Regression, and Boosted Trees, my recommendation to move forward in test implementation is Boosted Trees. Specific model measures are kept in the table below for reference. Both Logistical Regress and Boosted Trees improved AUC at 10 percentage points. Boosted trees performed slightly better than Logistical Regression, earning its spot for front runner for testing implementation. However, in evaluating significance between Logistical Regression and Boosted Trees, it was realized they did cross 0 and that there was no significant difference between the two models. Because of this, if during testing, the Boosted Trees Models is not performing well with realized churn, Logistical Regression would be a reasonable alternative model for testing implementation as well.

Method	Final-test AUC	ΔAUC vs. contract rule	95% interval for ΔAUC	Carry forward?	Comment
Contract rule	0.7373	—	—	Benchmark	Use information for benchmarking against current process
Logistic regression	0.8471	+0.1099	[0.0931, 0.1285]	Candidate	Has significant improvement against benchmarking. Good alternative to selected once in testing
Boosted trees	0.8497	+0.1124	[0.0959, 0.1311]	Selected	Has the highest improvement of AUC from the benchmark process. Selected to move forward into to testing

For implementation, I would suggest a 30 day testing period to measure the actual impact of the model. I would suggest piloting the new list with 20%-30% of your outreach team, based on risk tolerance and budget. For understanding business value, we use the cost of outreach ($6.20) and the net value of retention ($66). Using these numbers, we saw a benchmark breakeven save rate of about 23.5%. Utilizing either model, the breakeven dropped 10 percentage points to 13.5% and improved the financial case for testing. Cost savings and profitability are reliant on actual save rates upon model implementation. After the 30 day testing period, I would compare the actual save rate and profitability to the 13.5% breakeven threshold. If the actual save rate falls below that level, I would reconsider scaling and investigate model performance. Additionally, because Logistical Regression performed similarly, it could be then tested as an alternative. GitHub Repo Link for indepth project analysis.